

Maggie (Wanlin) Tu

513-764-2999 | maggietu570@gmail.com | www.linkedin.com/in/maggie-tu

EDUCATION

Vanderbilt University – Data Science Institute

M.S. Data Science

Nashville, TN

May 2026

University of Cincinnati

B.S. Business Analytics

Cincinnati, OH

December 2023

Chengdu University of Technology

B.S. Finance (Investment Banking Concentration), Transfer Coursework

Chengdu, China

July 2020

EXPERIENCE

FusionSite Services

Nashville, TN

Data Engineer Intern

May 2025 - Aug 2025

- Developed Azure Function App pipelines and PostgreSQL workflows to automate data ingestion, transformation, and reporting across safety, HR, and training systems.
- Analyzed 15,000+ driver safety events across 56 operational sites, identifying repeat-risk patterns and producing recommendations for regional leadership.
- Evaluated the effectiveness of a company-wide driver improvement program by analyzing 14,000+ cleaned records, uncovering performance trends among 250+ enrolled drivers.
- Built executive Power BI dashboards tracking headcount, turnover, training completion, and workforce KPIs, reducing manual reporting and improving visibility into organizational performance.
- Integrated DigitalChalk LMS training data into reporting workflows, enabling site-level tracking of compliance, completion rates, and employee engagement metrics.

Protiviti

Cincinnati, OH

Risk Analyst Intern

March 2024 - June 2024

- Conducted AML investigations on 50+ accounts per week, analyzing customer behavior, transaction activity, and KYC documentation to identify potential financial crime risks.
- Produced 200+ investigative reports and risk narratives, maintaining high quality standards while completing an average of 8 investigations daily.
- Reviewed 10,000+ transactions for indicators of money laundering, fraud, sanctions violations, and other compliance concerns.
- Collaborated with compliance and risk teams on case escalations, regulatory reporting, and financial crime monitoring.

WaterFire Providence

Providence, Rhode Island

Data Analyst Intern

June 2023 - August 2023

- Evaluated over 200 campaigns using R and Tableau to identify leading revenue generators over the past 3 years.
- Analyzed over 8000 donors, leveraging Python and Excel to identify lapsed donors from the past 5 years for a fundraiser event.
- Detected data redundancies and improved the clarity of fundraising campaign definitions to enhance organization's data management infrastructure.

PROJECTS

LLM-Powered Financial Signal Extraction from SEC Filings

- Analyzed 2,441 SEC filings spanning 2015–2025 and engineered a corpus of 67K+ financial sentiment factors using LLM-based reasoning and thematic factor extraction.
- Built financial NLP workflows leveraging DeepSeek-R1-Distill-Qwen-14B, QLoRA fine-tuning, and GRPO reinforcement learning to classify filing sentiment across five ordinal categories.
- Built a 5,217-sample annotation pipeline and evaluated model performance using Macro-F1 and portfolio-level alpha metrics.
- Designed event-driven backtests linking SEC disclosure sentiment to subsequent stock performance, generating investable signals across multiple holding horizons and sector cohorts.

Large-Scale Mental Health NLP Research (HLAB, Vanderbilt)

- Built scalable NLP workflows to analyze 918K+ Reddit submissions using DLATK, LDA topic modeling, logistic regression, and statistical significance testing.
- Engineered user-level feature tables and generated 100-topic latent semantic representations to compare language patterns across three mental health communities.
- Applied Differential Language Analysis (DLA) to 50K+ n-gram features, identifying clinically interpretable language markers associated with PTSD, cPTSD, and anxiety.
- Analyzed longitudinal trends in trauma-related discourse and support-seeking behaviors from 2020–2025.

SKILLS

Python, SQL, R, PySpark, Snowflake, Databricks, Azure

Machine Learning, NLP, LLMs, QLoRA Fine-Tuning, DeepSeek-R1, GPT-4o

Power BI, Tableau, Git, MySQL

English, Mandarin Chinese (Native)